

Tag 4

Guten Morgen!



Azure AI & Copilot

Microsoft Applied Skills

Sie entwickeln praxisnah KI-Anwendungen, Sprach- und Bildanalysen in Azure sowie eigene Copilots – von RAG bis Knowledge Mining. Für Entwickler*innen und Data Scientists, die innovative KI-Lösungen mit Azure AI-Services professionell einsetzen wollen.

Thomas Jäkel



Lead Trainer Cloud Infrastructure
Microsoft Certified Trainer since 1999

<https://github.com/www42/azkc>



Microsoft Certified:
Azure AI Fundamentals





Applied Skills credential earned

Microsoft Applied Skills: Build a generative AI chat app

2025-06-11 10:00:00

PASS

Assessment date: June 11, 2026

Your overall results: **76%** Pass

65% needed to pass



Performance by task



Prepare and deploy a model in Microsoft Foundry



Create a chat app by using Microsoft Foundry SDK



Implement chat history

gpt4-mini + Custom Guardrails

client = OpenAI() X

Marcel MAX_TURNS = 10 ...



Microsoft Applied Skills

Learn more at: learn.microsoft.com/credentials

Stand 11.06.2026

Don't just know it. Show it. Demonstrate your ability to use Microsoft technologies in real-world AI and cloud scenarios. Choose from technical and new **business-focused credentials**.

Technical
Business



Beginner

Cloud & AI Platforms

Create an AI agent

Get started with classes, properties, and methods in C#

Get started with cloud security and monitoring tasks

Get started with Azure management tasks

New

Get started developing agents in Microsoft Foundry

AI Business Solutions

Generate reports by using AI research agents

Streamline business workflows with AI chat

Security

Get started with identities and access using Microsoft Entra

Intermediate

Accelerate app development by using GitHub Copilot

Configure secure access to your workloads using Azure networking

Implement a Real-Time Intelligence solution with Microsoft Fabric

New

Resolve GitHub issues by using GitHub Copilot

Automate Azure Load Testing by using GitHub Actions

Create and manage canvas apps with Power Apps

Defend against cyberthreats with Microsoft Defender XDR

Secure AI solutions in the cloud

Administer Active Directory Domain Services

Deploy and configure Azure Monitor

Implement a data warehouse in Microsoft Fabric

New

Integrate model context protocol tools with agents in Microsoft Foundry

Create agents in Microsoft Copilot Studio

Create and manage model-driven apps with Power Apps and Dataverse

Implement information protection and data loss prevention by using Microsoft Purview

Secure Azure services and workloads with Microsoft Defender for Cloud regulatory compliance controls

Build a natural language processing solution with Azure AI Language

Deploy cloud-native apps using Azure Container Apps

Implement security through a pipeline using Azure DevOps

Create and manage automated processes by using Power Automate

Prepare security and compliance to support Copilot for Microsoft 365

Implement retention, eDiscovery, and Communication Compliance in Microsoft Purview

Coming soon

Protect information in Microsoft 365 Copilot by using Microsoft Purview

Build AI-powered solutions by using Microsoft Azure Database for PostgreSQL

Develop data-driven applications by using Microsoft Azure SQL Database

Migrate SQL Server workloads to Azure SQL Database

Build a generative AI chat app

Enhance agents with autonomous capabilities

Secure storage for Azure Files and Azure Blob Storage



New Microsoft Certification exams (beta) timeline

March

April

May

June

Machine Learning Operation
(MLOps) Engineer Associate

AI-300

Data

Azure AI App and Agent
Developer Associate

AI-103

Developer

Azure AI Cloud Developer
Associate

AI-200

Developer

Windows Server Hybrid
Administrator

AZ-802

IT Pro

Azure Databricks Data
Engineer Associate

DP-750

Data

Azure AI Fundamentals

AI-901

IT Pro

Cloud and AI Security Engineer
Associate

SC-500

Security Pro

SQL AI Developer Associate

DP-800

Developer

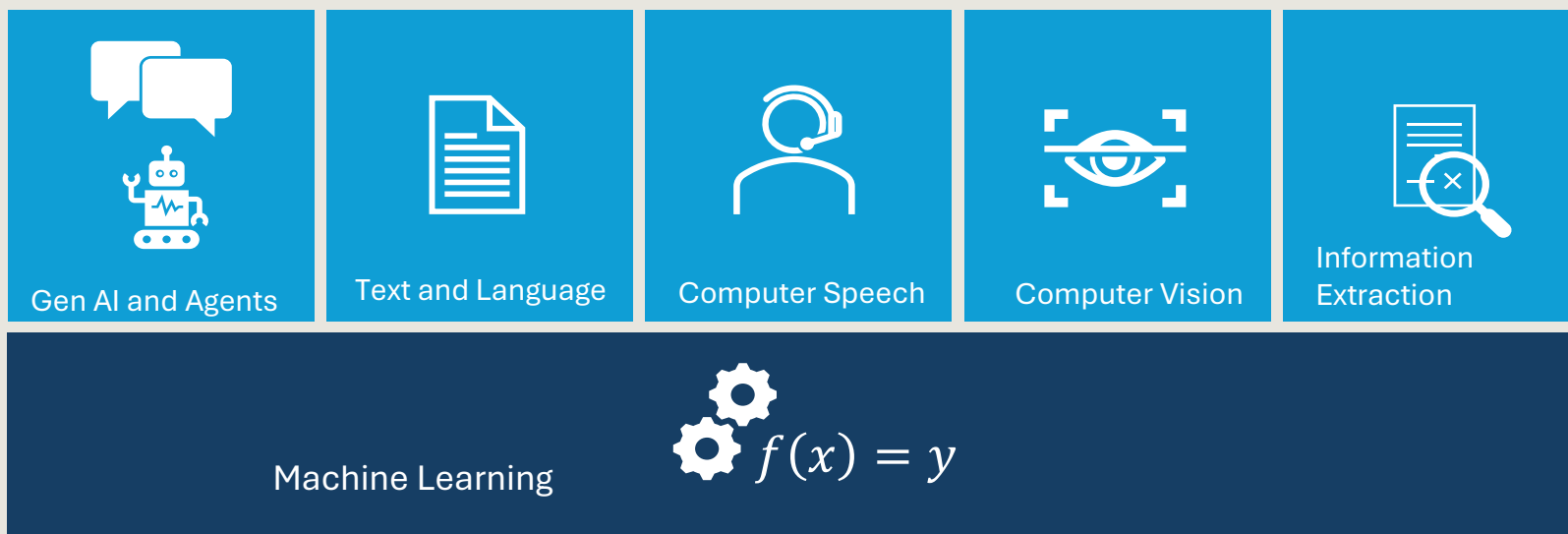
Cybersecurity Business
Professional

SC-730

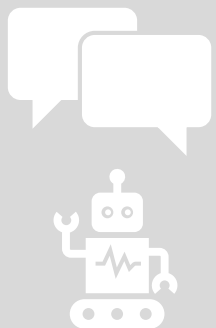
Business Pro

Agenda

1. Foundry, Models, Agents
2. VS Code, Python, Chat App
3. Model Context Protocol MCP
4. Language, Speech, Vision, Content Understanding



Our focus today



Generative AI and Agents



Text and Language



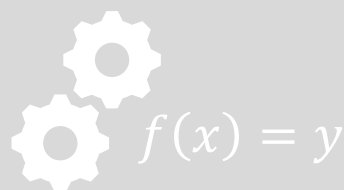
Computer Speech



Computer Vision



Information Extraction



Machine Learning



Develop natural language solutions in Azure

<https://aka.ms/mslearn-language>



Analyze text with Azure Language in Foundry Tools

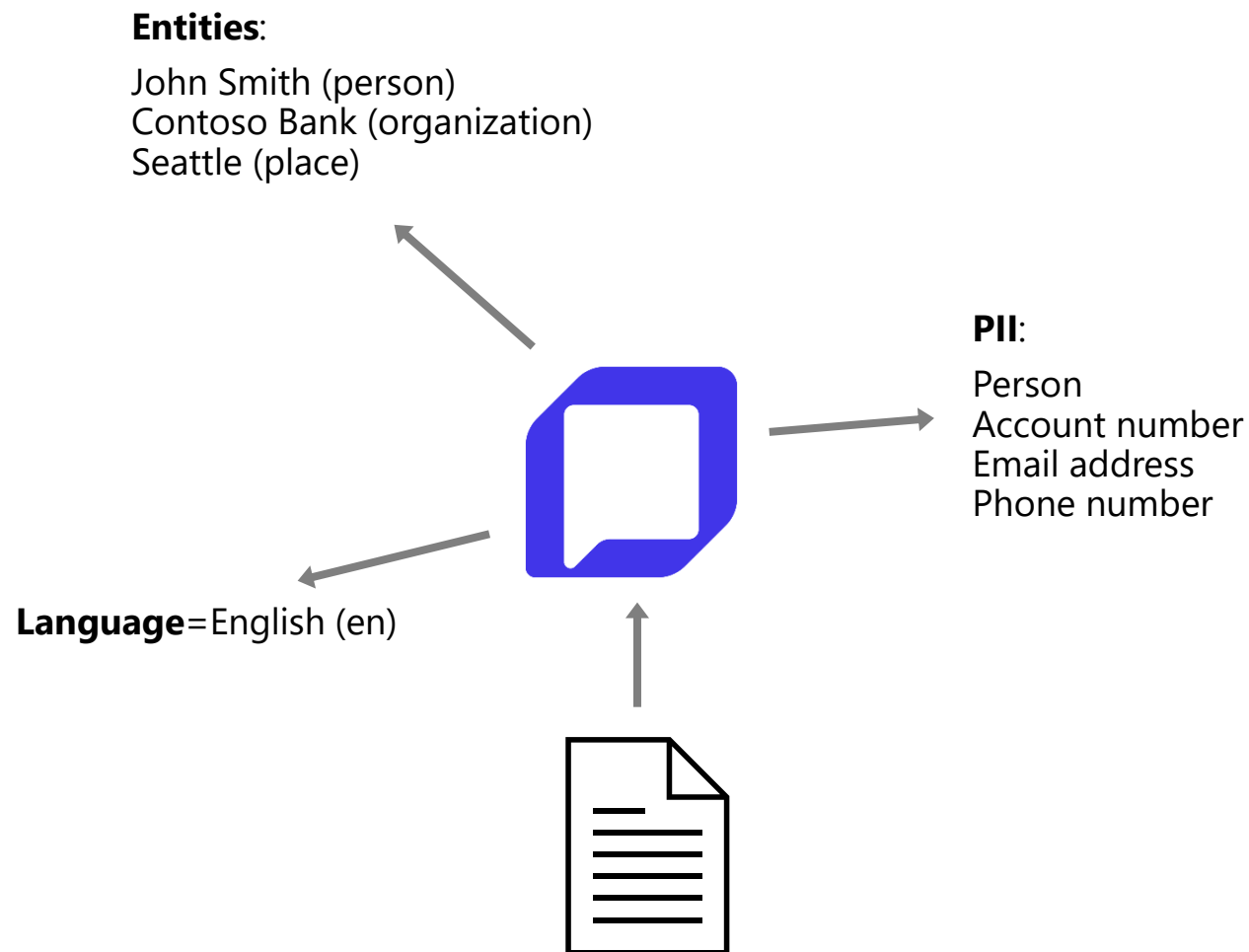
<https://aka.ms/mslearn-analyze-text>



Azure Language in Foundry Tools

Prebuilt models for:

- Language detection
- Named entity recognition
- PII detection



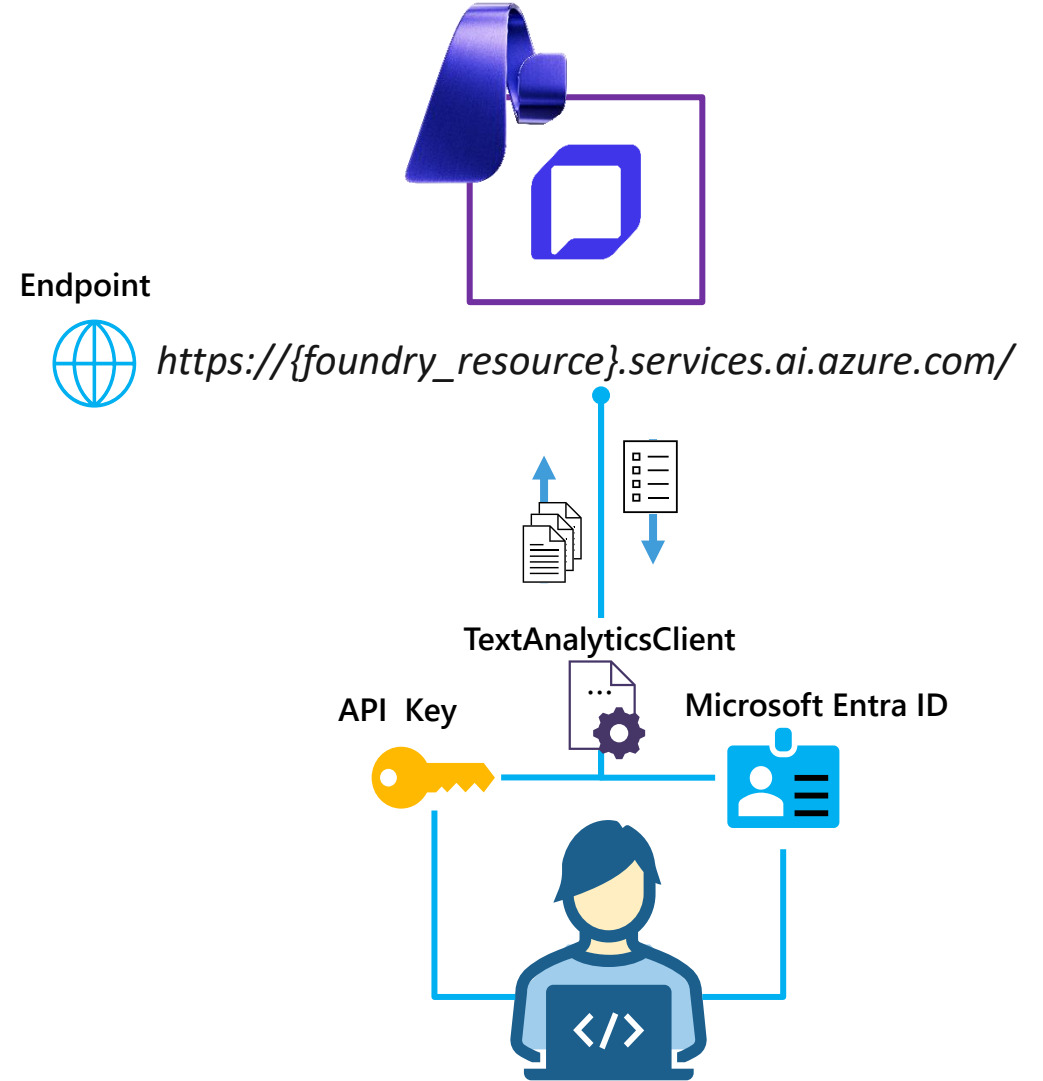
Using the Text Analytics API

1. Connect to the API

- **Endpoint:**
 - Microsoft Foundry resource*
- **Credentials:**
 - Microsoft Foundry project API key
 - Microsoft Entra ID

2. Use a *TextAnalyticsClient* to call the appropriate function for your task

- **Request:** One or more text documents
- **Response:** Task-specific collection of results (one per document)



Exercise

Analyze text

In this exercise, you'll use Azure Language to:

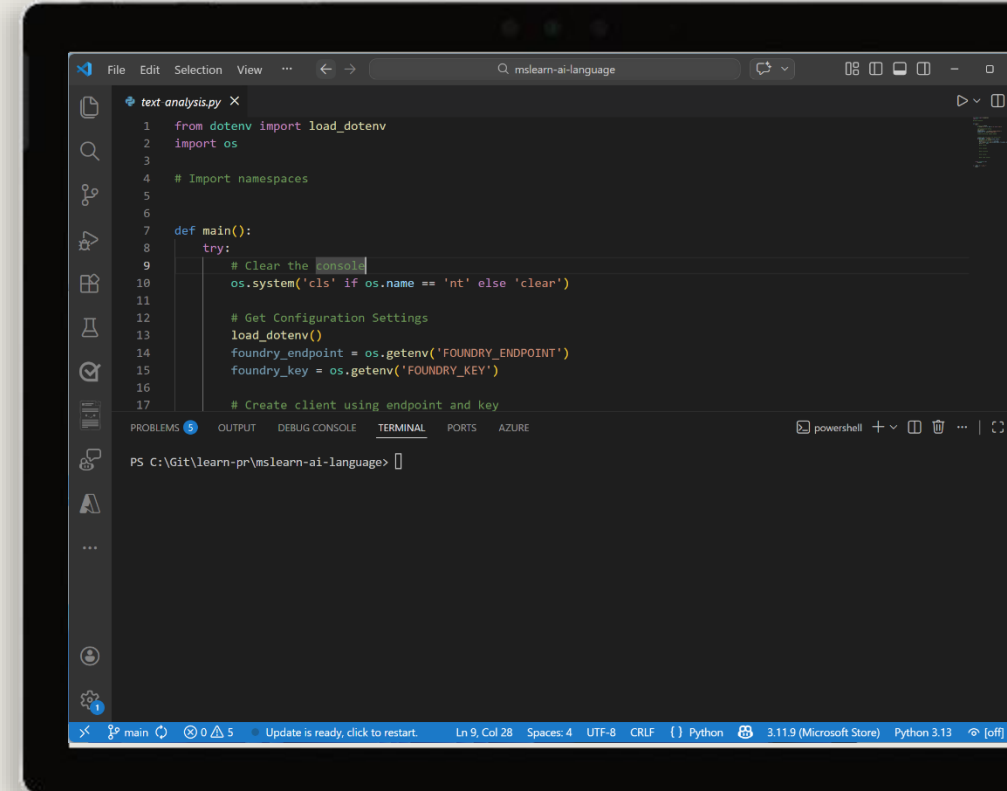
- Detect Language
- Extract Entities
- Extract PII

Start the exercise at:

13. Analyze text

🕒 1 h

0%



Knowledge check



1 How should you create an application that analyzes news articles and extracts key people, places, and dates that are mentioned for indexing?

- ☐ Use a generative AI model with a custom function tool that matches strings using a regular expression.
- ☐ Use Azure Language in Foundry Tools to extract PII entities.
- ☒ Use Azure Language in Foundry Tools to extract named entities.

2 You want to publish extracts from customer testimonials on a web site. You need to remove personal details from the text before publishing it. What should you do?

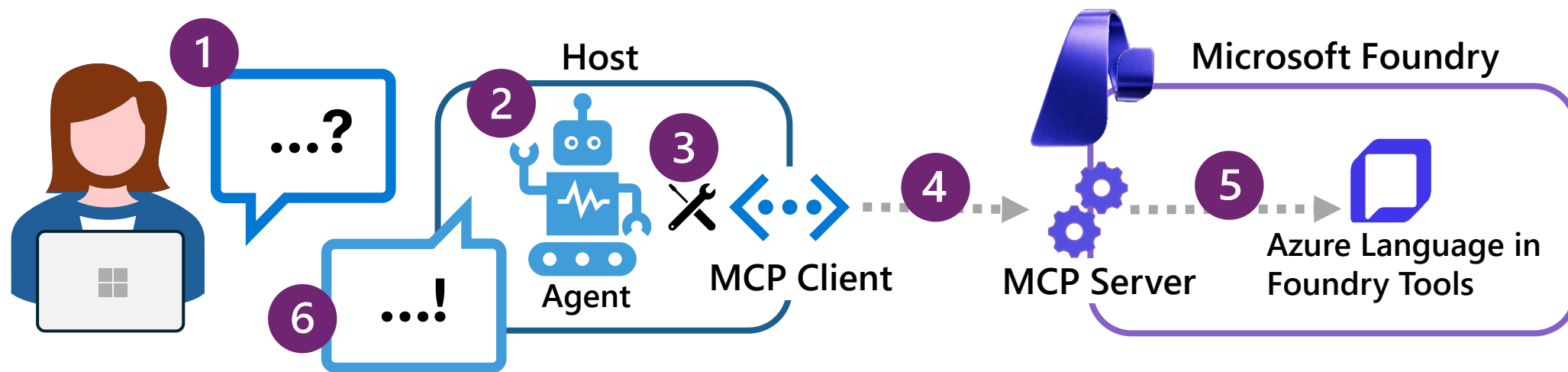
- ☒ Use Azure Language in Foundry Tools to find and redact PII entities.
- ☐ Use Azure Language in Foundry Tools to detect the language and publish only the testimonials in English.
- ☐ Use a gpt-4.1 model to create new AI-generated customer reviews.

Develop a text analysis agent with the Azure Language MCP server

<https://aka.ms/mslearn-language-mcp>



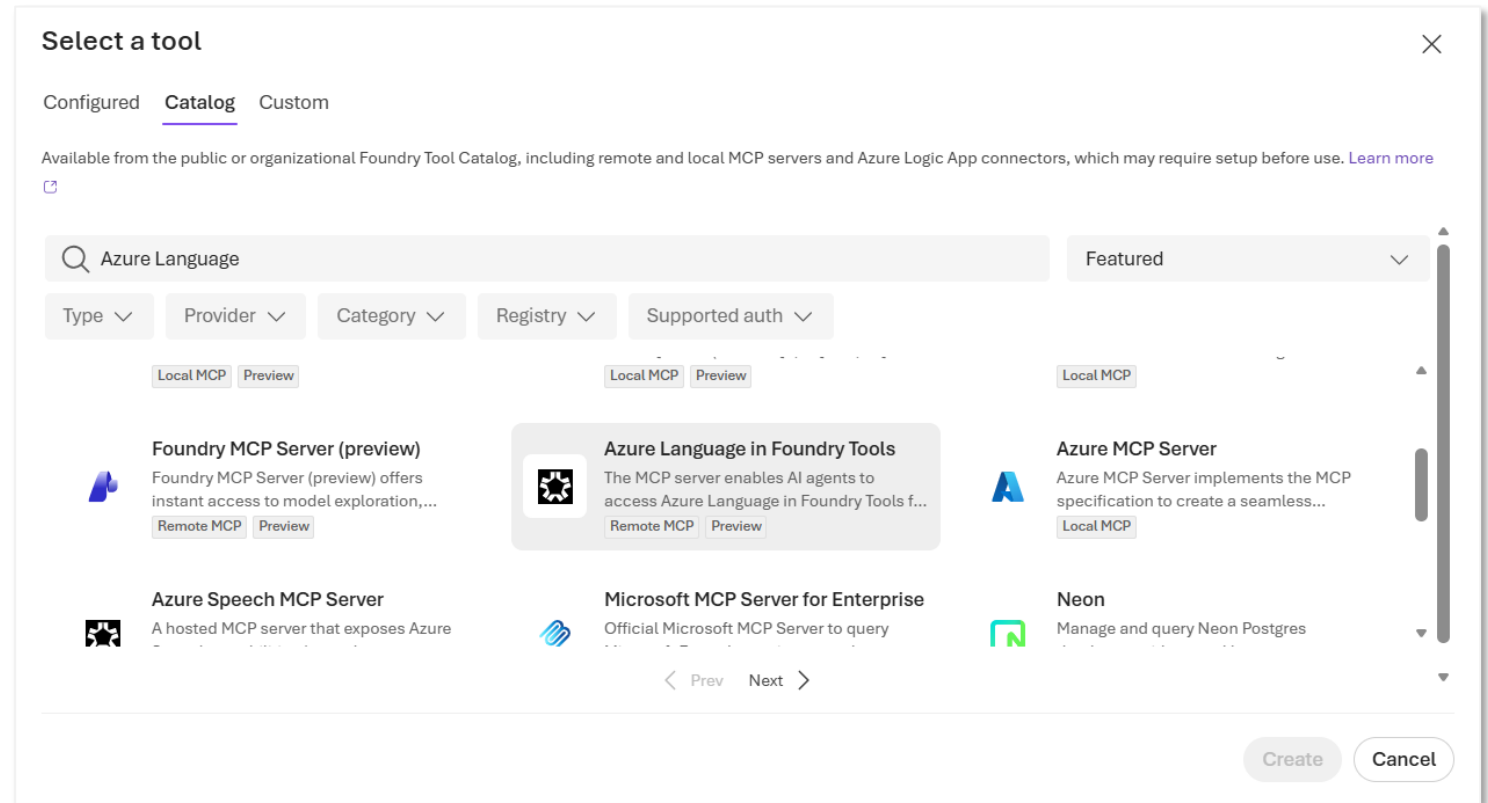
The Azure Language MCP server



1. A user sends a prompt to an agent
2. The agent determines task (or tasks) to be performed
3. The agent checks available MCP tools to find the best match
4. The agent calls the selected tool through the MCP server, passing the relevant input text
5. The MCP server processes the request using the appropriate Azure Language capability and returns results
6. The agent combines the results into a natural language response for the user

Using the Azure Language MCP server

1. Create an agent
2. Add the Azure Language MCP Server tool
3. Specify agent instructions to select the tool when appropriate
4. Test in the agent playground
5. Build a client application that uses the agent



Exercise

Develop a text analysis agent

In this exercise, you'll:

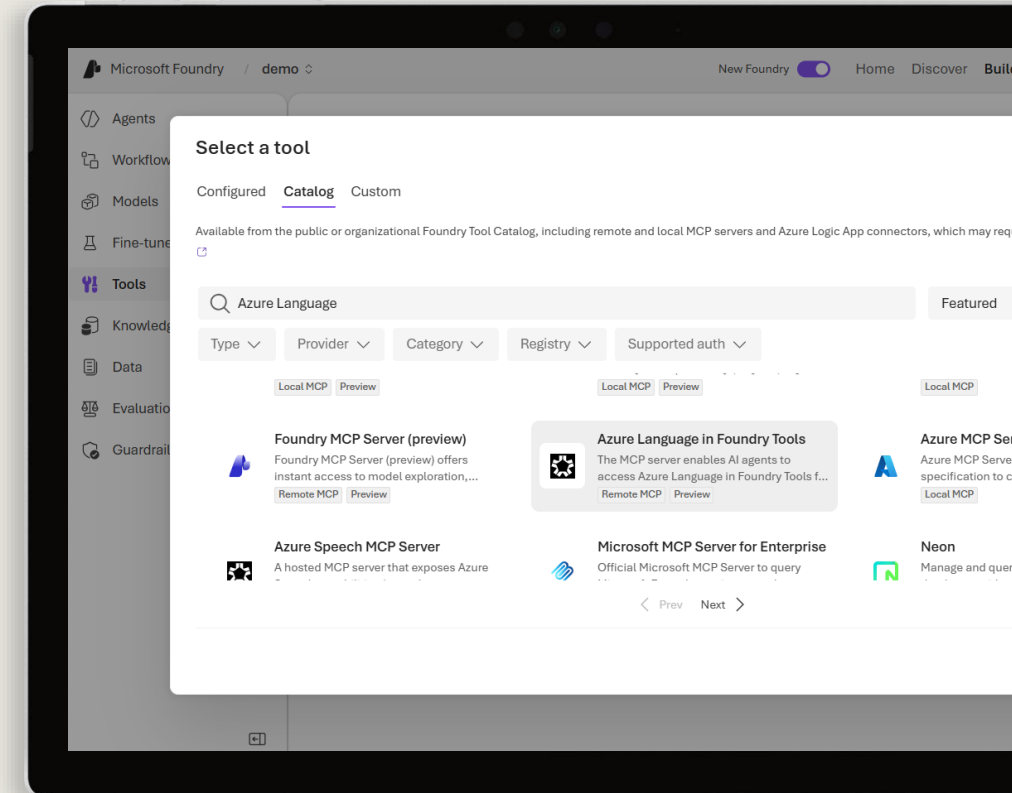
- Create an agent in Microsoft Foundry
- Add and test the Azure Language MCP tool
- Create a client application for the agent

Start the exercise at:

14. Develop a text analysis agent

🕒 1 h

0%



Knowledge check



1 What is the primary role of the Azure Language MCP server?

- ☐ To train and fine-tune custom language models for use by AI agents
- ☒ To expose Azure Language text analysis capabilities as MCP tools for agents
- ☐ To deploy and manage large language models in an Azure subscription

2 How does an agent determine which Azure Language MCP tool to call when processing a user's prompt?

- ☐ The developer writes routing logic to direct each prompt to a specific tool
- ☒ The agent matches the prompt to tool descriptions received from the MCP server
- ☐ The MCP server analyzes the prompt and automatically routes it to a tool

Develop a speech-capable generative AI application

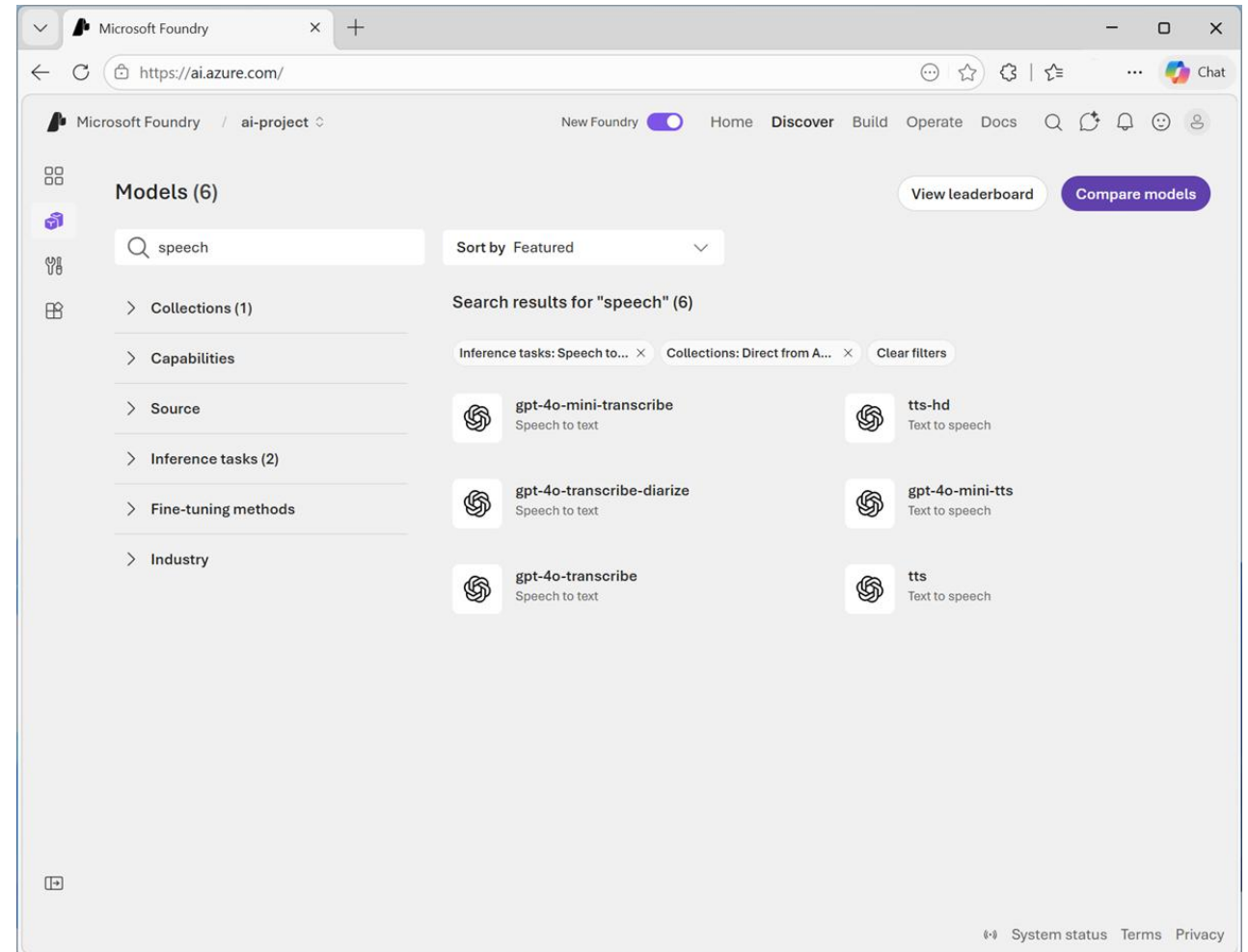
<https://aka.ms/mslearn-generative-ai-speech>



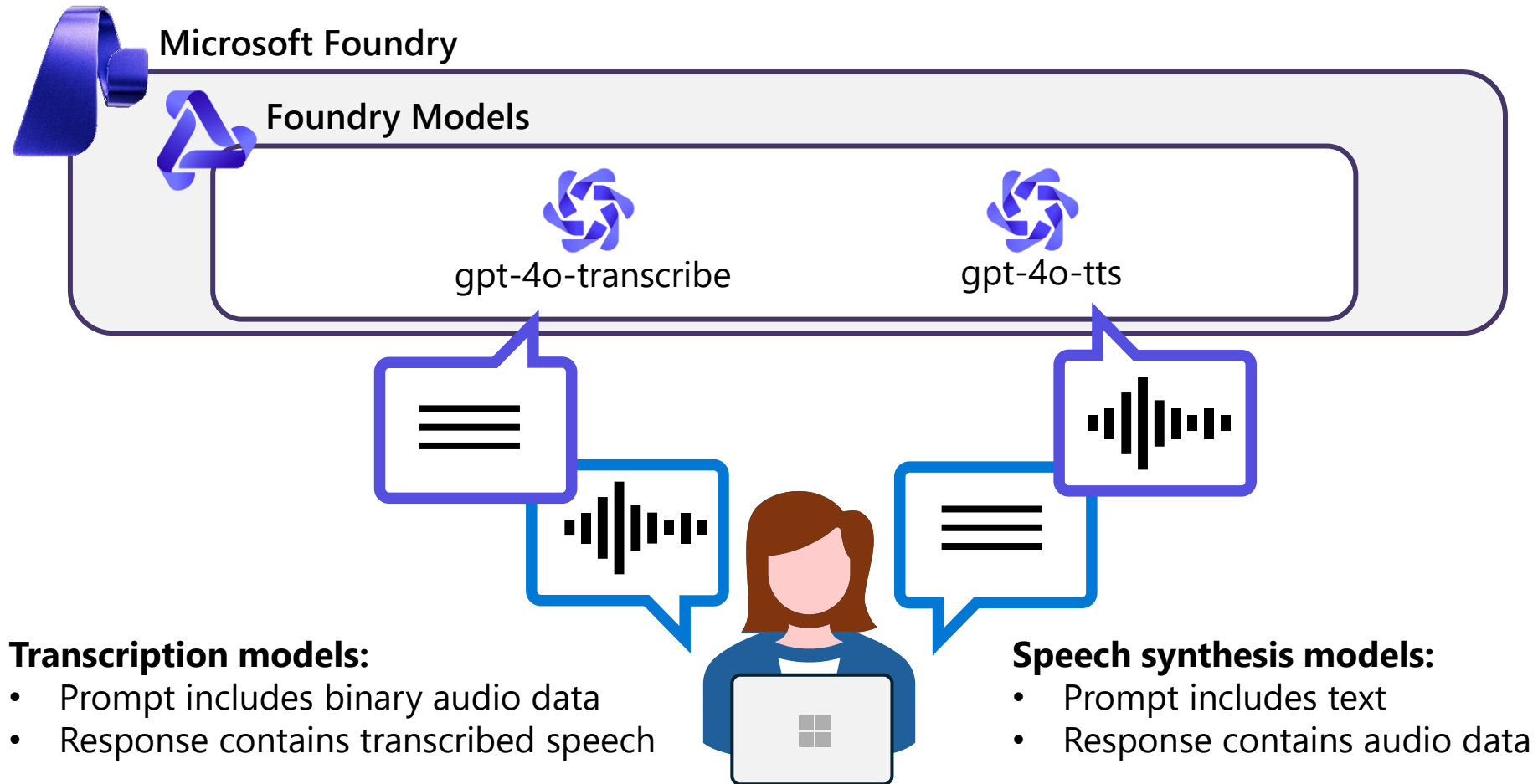
Speech-capable models in Microsoft Foundry Models

Filter and search for:

- Generative AI models that can transcribe speech to text
- Generative AI models that can synthesize text to speech



Using speech-capable generative AI models



Exercise

Use speech-capable generative AI models

In this exercise, you'll:

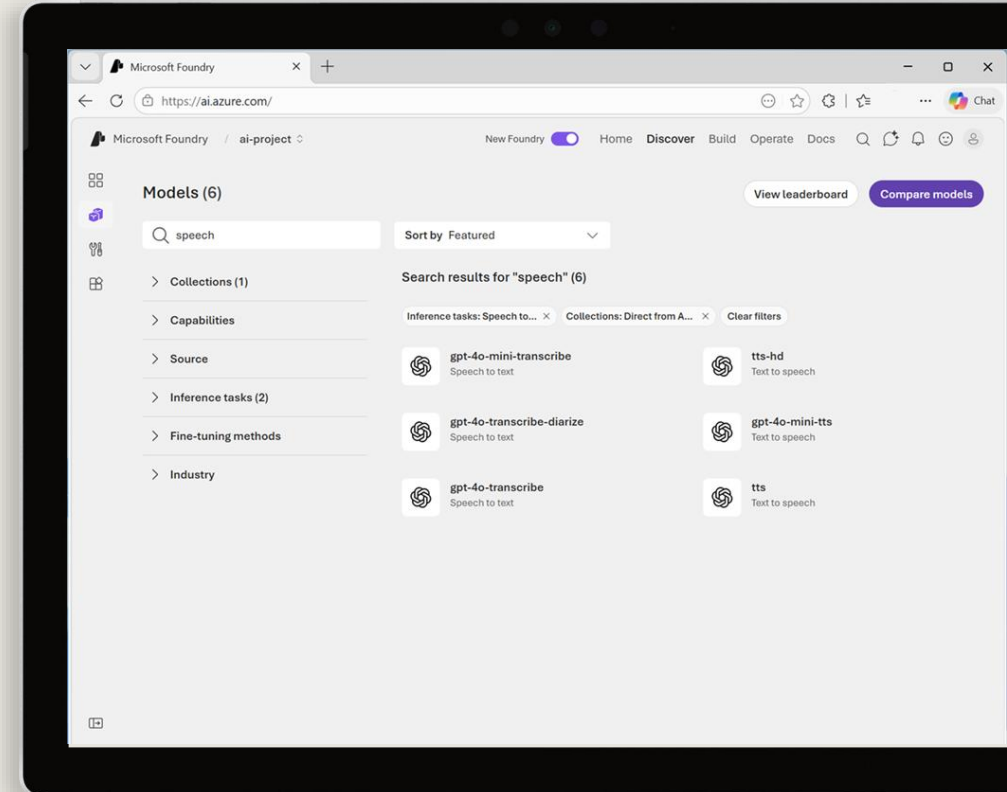
- Find and deploy speech-capable models in Microsoft Foundry
- Create a speech synthesis app
- Create a speech transcription app

Start the exercise at:

15. Use speech-capable generative AI models

🕒 1 h

0%



Knowledge check



1 Which model can you use to generate text from speech?

- ☐ gpt-4o-mini
- ☐ gpt-4o-mini-tts
- ☒ gpt-4o-mini-transcribe

2 Which model can you use to synthesize speech from text?

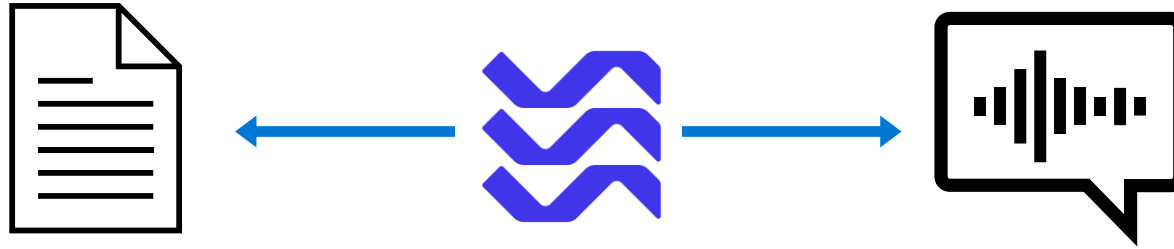
- ☐ gpt-4o-mini
- ☒ gpt-4o-mini-tts
- ☐ gpt-4o-mini-transcribe

Create speech-enabled apps with Microsoft Foundry

<https://aka.ms/mslearn-speech-app>



Azure Speech in Foundry Tools



- **Speech to text:** Transcribe spoken audio
- **Text to speech:** Synthesize speech
- **Voice Live:** Real-time conversational speech interactions
- **Speech translation:** Multilingual translation

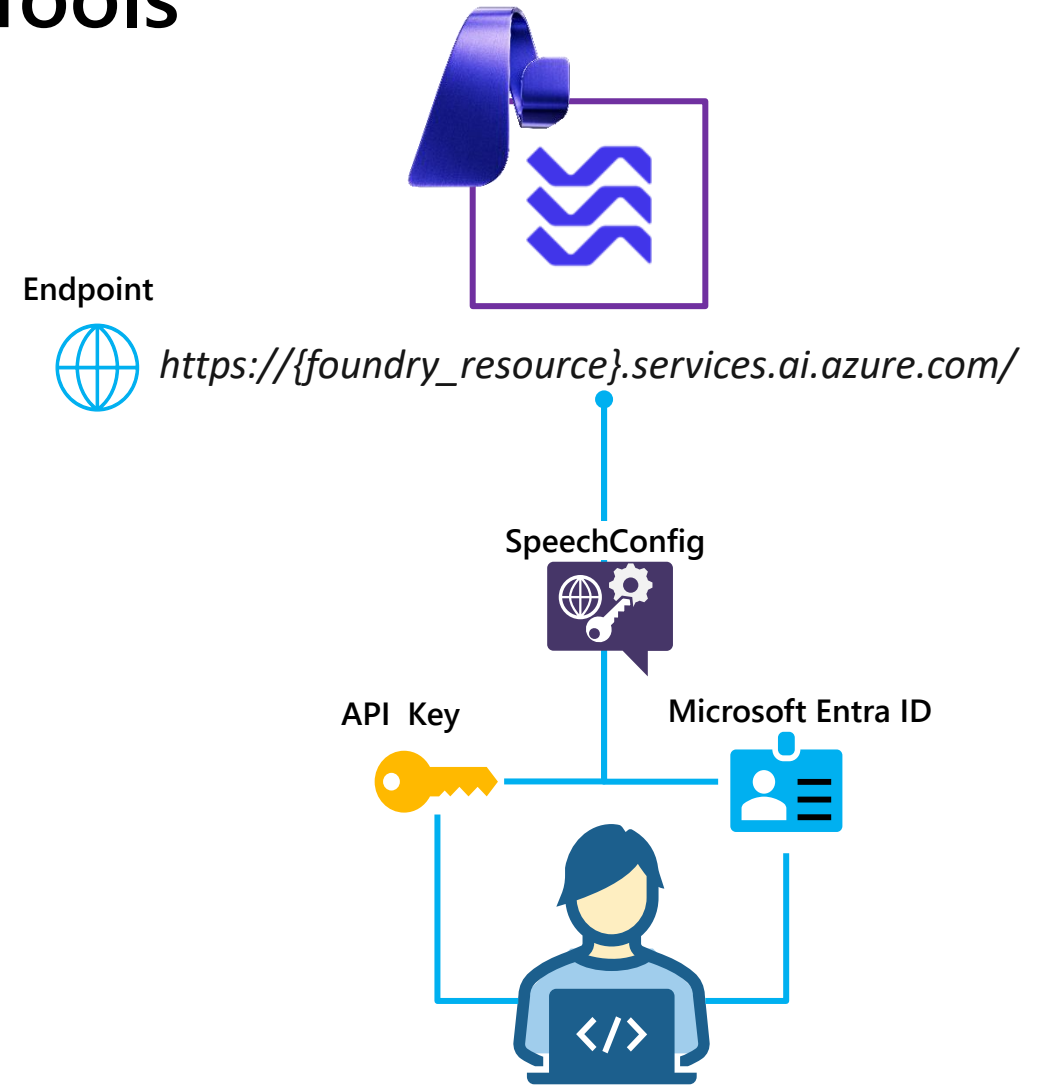
Using Azure Speech in Foundry Tools

1. Connect to the API

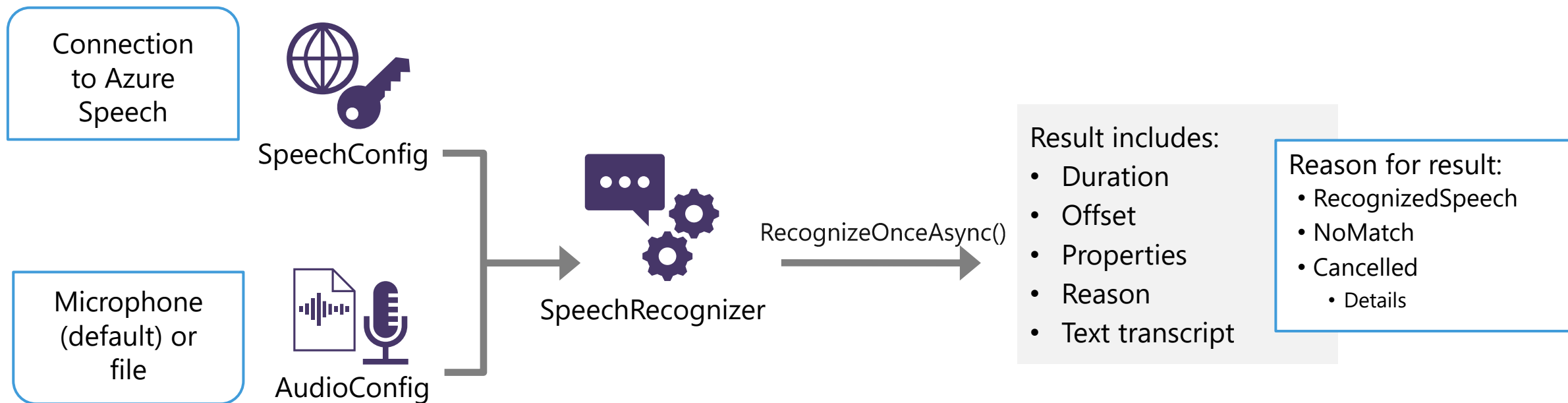
- **Endpoint:**
 - Microsoft Foundry resource*
 - Global/Regional endpoint
- **Credentials:**
 - Microsoft Foundry project API key
 - Microsoft Entra ID

2. Create a *SpeechConfig* object

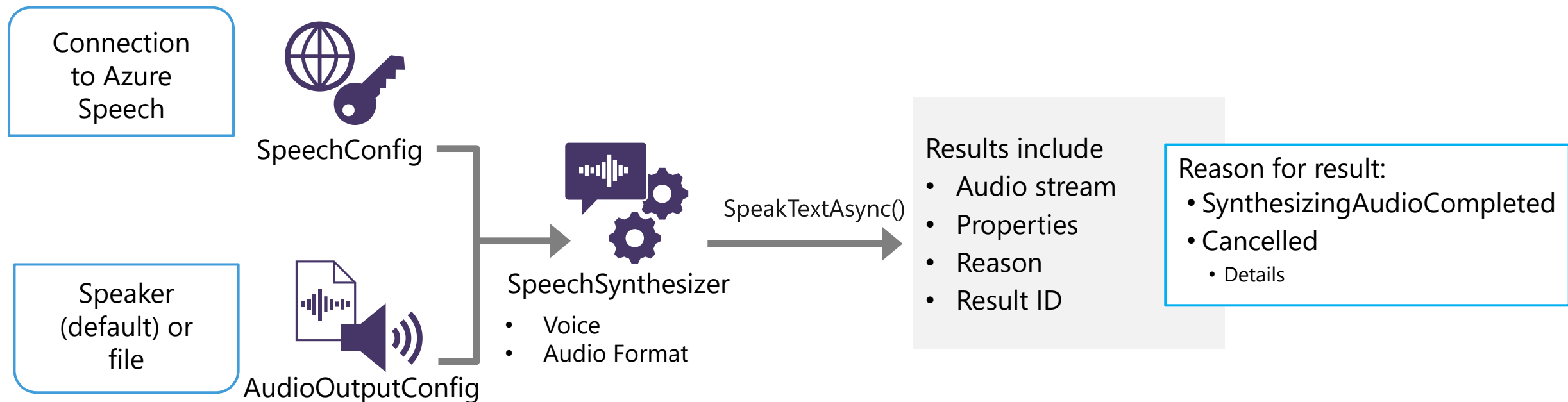
- Proxy object for speech API operations



Speech to text



Text to speech



Speech Synthesis Markup Language (SSML)

XML-based language with customization options:

- Speaking styles (Neural voices only)
- Pauses and silence
- Phonemes (phonetic pronunciations)
- Prosody (speaking pitch, range, rate, etc.)
- "say-as" (number, date, time, address, etc.)
- Insert recorded speech or background audio

```
SpeakSsmlAsync( ssml-string );
```

```
<speak version="1.0" xmlns="http://www.w3.org/2001/10/synthesis"
        xmlns:mstts="https://www.w3.org/2001/mstts" xml:lang="en-US">
  <voice name="en-US-AriaNeural">
    <mstts:express-as style="cheerful">
      I say tomato
    </mstts:express-as>
  </voice>
  <voice name="en-US-GuyNeural">
    I say <phoneme alphabet="sapi" ph="t ao m ae t ow"> tomato </phoneme>.
    <break strength="weak"/> Lets call the whole thing off!
  </voice>
</speak>
```

Multiple voices in a single synthesis

Speaking style

Phonetic pronunciation

Pause

Exercise

Recognize and synthesize speech

In this exercise, you'll use Azure Speech to build an app that can:

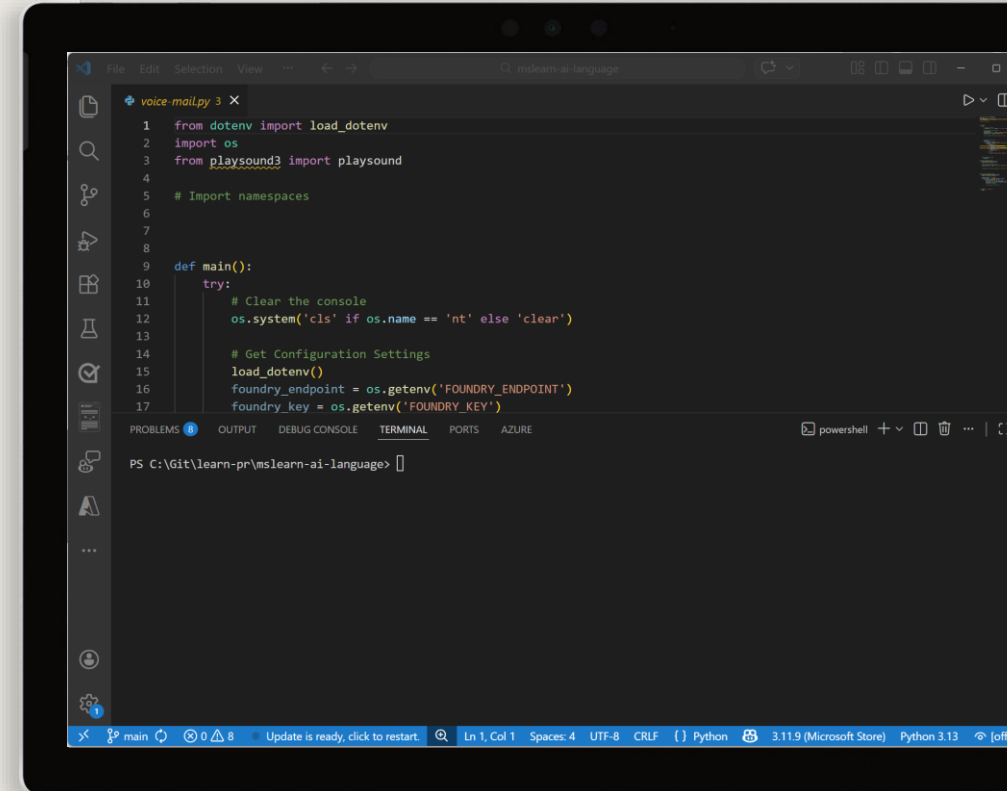
- Synthesize speech
- Recognize speech

Start the exercise at:

16. Recognize and synthesize speech

🕒 No limit

0%



Knowledge check



1 What information do you need from your Microsoft Foundry resource to consume it using the Azure Speech SDK?

- ☒ The endpoint and key
- ☐ The primary and secondary keys
- ☐ The Azure subscription ID and resource group name

2 Which object should you use to specify that the speech input to be transcribed to text is in an audio file?

- ☐ SpeechConfig
- ☒ AudioConfig
- ☐ SpeechRecognizer

3 How can you change the voice used in speech synthesis?

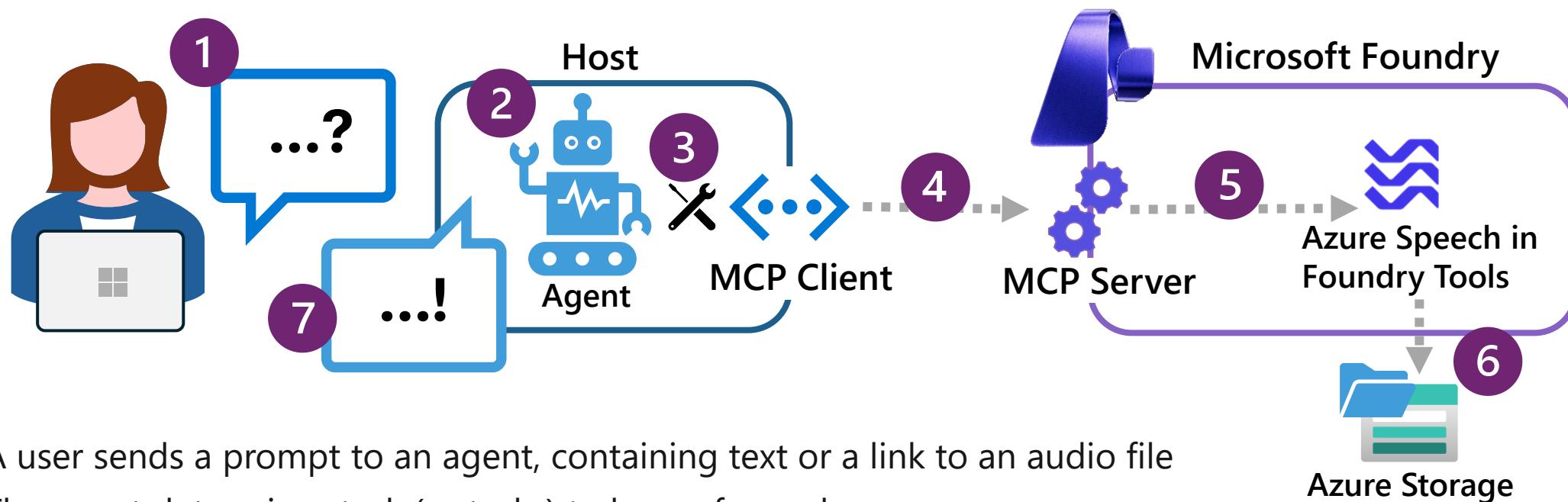
- ☐ Specify a SpeechSynthesisOutputFormat enumeration in the SpeechConfig object
- ☒ Set the SpeechSynthesisVoiceName property of the SpeechConfig object to the desired voice name
- ☐ Specify a filename in the AudioConfig object

Develop a speech agent with the Azure Speech MCP server

<https://aka.ms/mslearn-speech-mcp>



The Azure Speech MCP server



1. A user sends a prompt to an agent, containing text or a link to an audio file
2. The agent determines task (or tasks) to be performed
3. The agent checks available MCP tools to find the best match
4. The agent calls the selected tool through the MCP server, passing the relevant input
5. The MCP server processes the request using Azure Speech (text to speech or speech to text)
6. In the case of text to speech, the generated audio is saved as a file in Azure storage
7. The agent responds to the user with transcribed text or a link to the generated audio file

Using the Azure Language MCP server

1. Create an agent
2. Connect the Azure Speech MCP Server tool

Requires an Azure Storage container
SAS URL

3. Specify agent instructions to select the tool when appropriate
4. Test in the agent playground
5. Build a client application that uses the agent

Connect the Azure Speech MCP Server tool

A Foundry connection will be created for you in your Foundry project based on the values you provided below.

Name *

Remote MCP Server endpoint *

Parameters *

:

Authentication *

Authorization *

Exercise

Use Azure Speech in an agent

In this exercise, you'll:

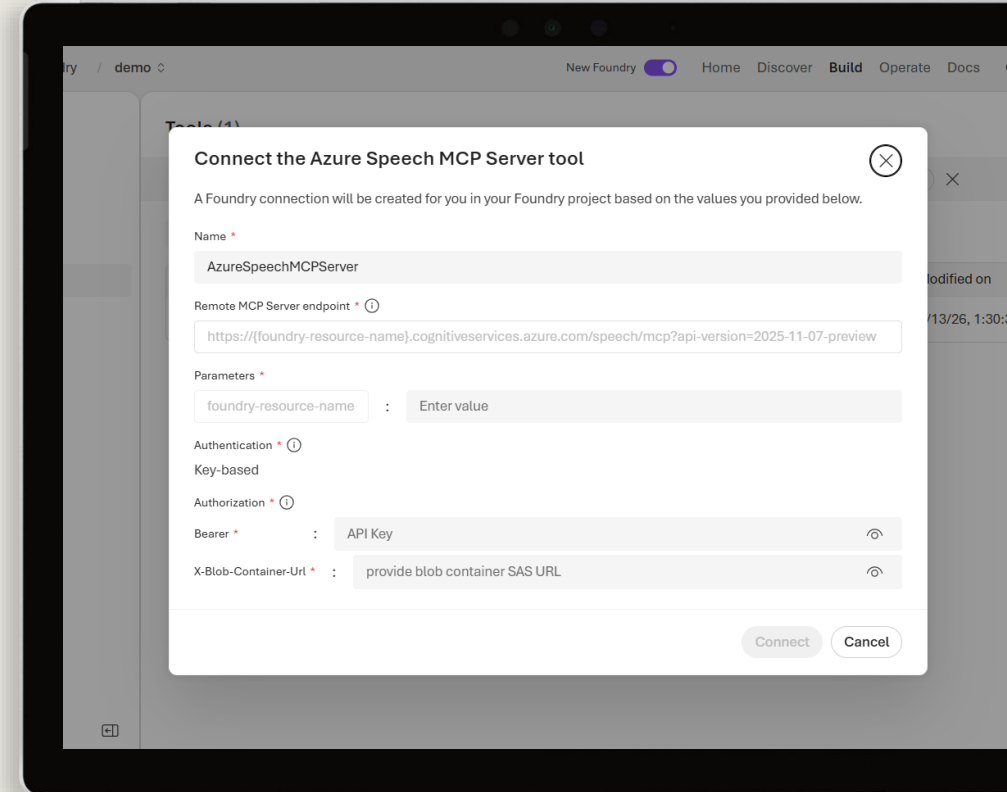
- Create an agent in Microsoft Foundry
- Add and test the Azure Speech MCP tool
- Create a client application for the agent

Start the exercise at:

17. Use Azure Speech in an agent

🕒 1 h

0%



Knowledge check



1 What two core capabilities does the Azure Speech MCP server expose to agents?

- ☐ Language translation and text summarization
- ☒ Speech-to-text recognition and text-to-speech synthesis
- ☐ Named entity recognition and sentiment analysis

2 Why does the Azure Speech MCP server require an Azure Storage account?

- ☐ To store the agent's instructions and configuration settings
- ☒ To store input audio files and output audio files generated by the speech tools
- ☐ To cache the MCP server's tool definitions for faster discovery

3 What credentials are needed when connecting the Azure Speech MCP server to a Foundry agent?

- ☐ An OAuth 2.0 token and a managed identity endpoint URL
- ☒ A Foundry resource key and a SAS URL for a blob container
- ☐ A client certificate and the Azure subscription ID

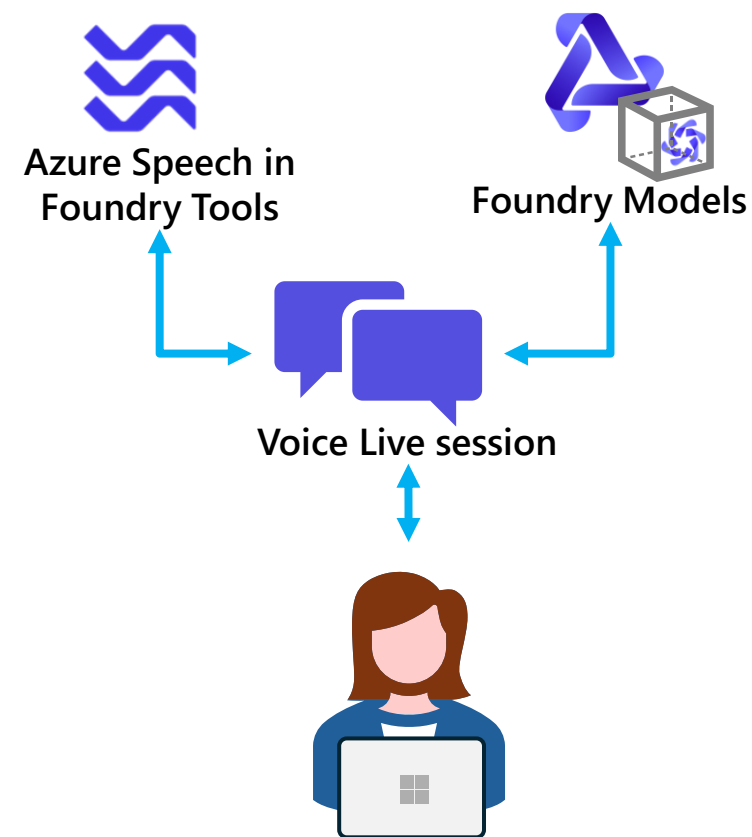
Develop an Azure Speech Voice Live Agent in Microsoft Foundry

<https://aka.ms/mslearn-voice-live-agent>



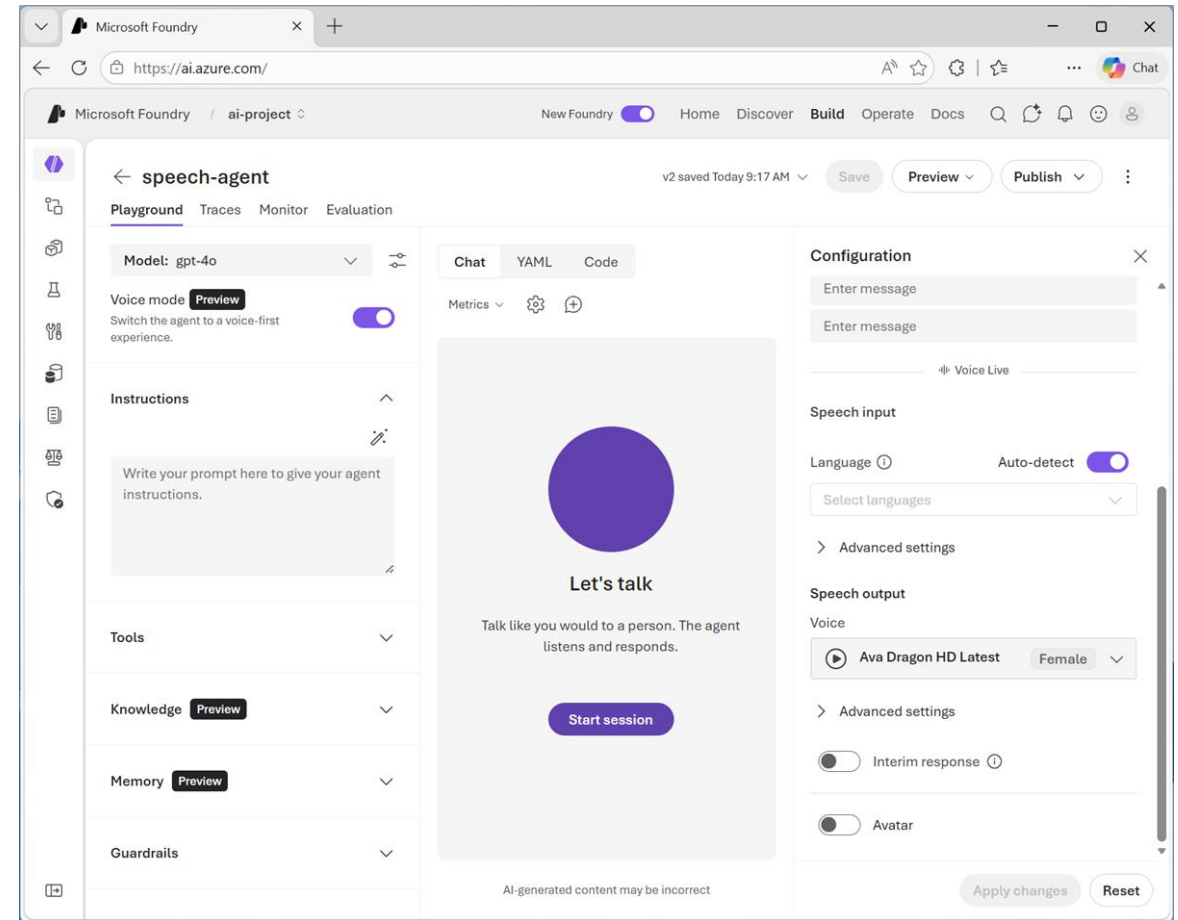
Azure Speech Voice Live

- Real-time, event-based conversation framework
- Connects models to speech conversations in a *Voice Live session*
- Authentication options are:
 - Microsoft Entra ID (keyless)
 - API key
- Advanced features include:
 - Real-time audio processing with support for multiple formats like PCM16 and G.711
 - Voice options, including OpenAI voices and Azure custom voices
 - Avatar integration using WebRTC for video and animation
 - Built-in noise reduction and echo cancellation



Creating a Voice Live agent

- Enable **Audio mode** in the agent playground
 - Configure *Voice Live* settings for the agent
- Create a client app that uses the Voice live SDK to:
 1. Connect to the agent
 2. Configure audio hardware input and output
 3. Establish a *Voice Live* session
 4. Monitor audio systems for activity
 5. Process events (such as user speech input and responses from the agent)



Exercise

Develop a Voice Live agent

In this exercise, you'll:

- Enable and configure Azure Speech Voice Live for an agent
- Implement a conversational app that uses the agent

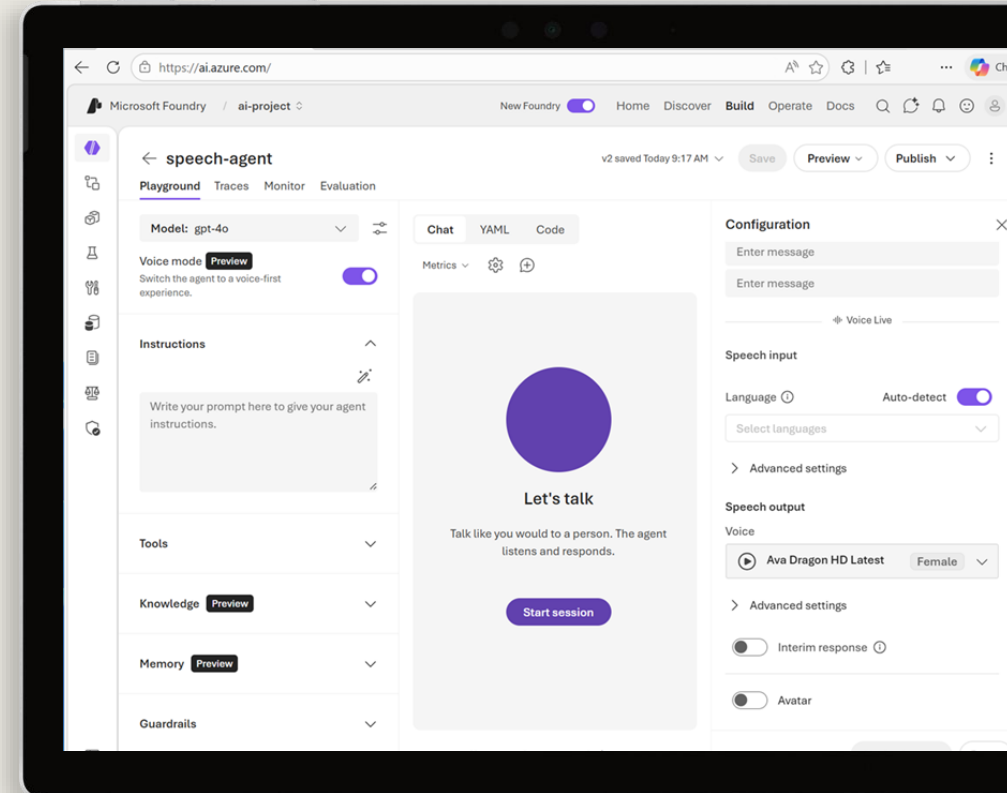
Start the exercise at:

18. Develop a Voice Live agent



1 h

0%



Knowledge check



1 What are the two authentication methods supported by the Voice Live API?

- ☐ OAuth 2.0 and JWT (JSON Web Tokens)
- ☐ Basic authentication and API keys
- ☒ Microsoft Entra (keyless) and API key

2 How do you configure and test Voice Live agent integration in the Foundry Portal?

- ☐ You can't – Voice Live is only accessible through the REST API or Python SDK
- ☐ In the Azure Speech in Foundry Tools Voice Live playground
- ☒ Enable **Voice mode** in the agent playground

3 How can you stop audio playback when a user interrupts the voice agent?

- ☐ You can't – the user must wait for the agent to finish
- ☒ Handle the `INPUT_AUDIO_BUFFER_SPEECH_STARTED` event
- ☐ Reset the Voice Live session and clear the conversation history

Translate text and speech with Microsoft Foundry Tools

<https://aka.ms/mslearn-translate>



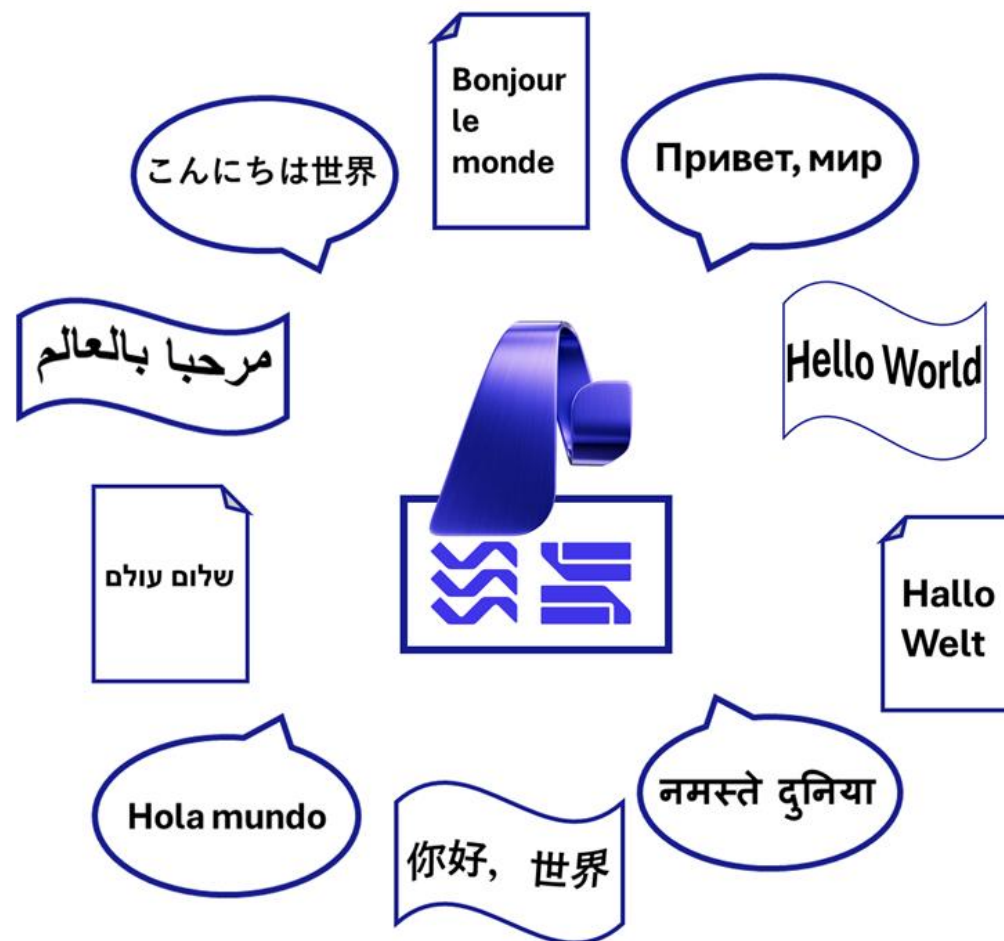
Translation in Microsoft Foundry

Azure Translator in Foundry Tools

- Multilingual translation for:
 - Text
 - Documents

Azure Speech in Foundry Tools

- Multilingual speech translation:
 - Speech to text
 - Speech to speech



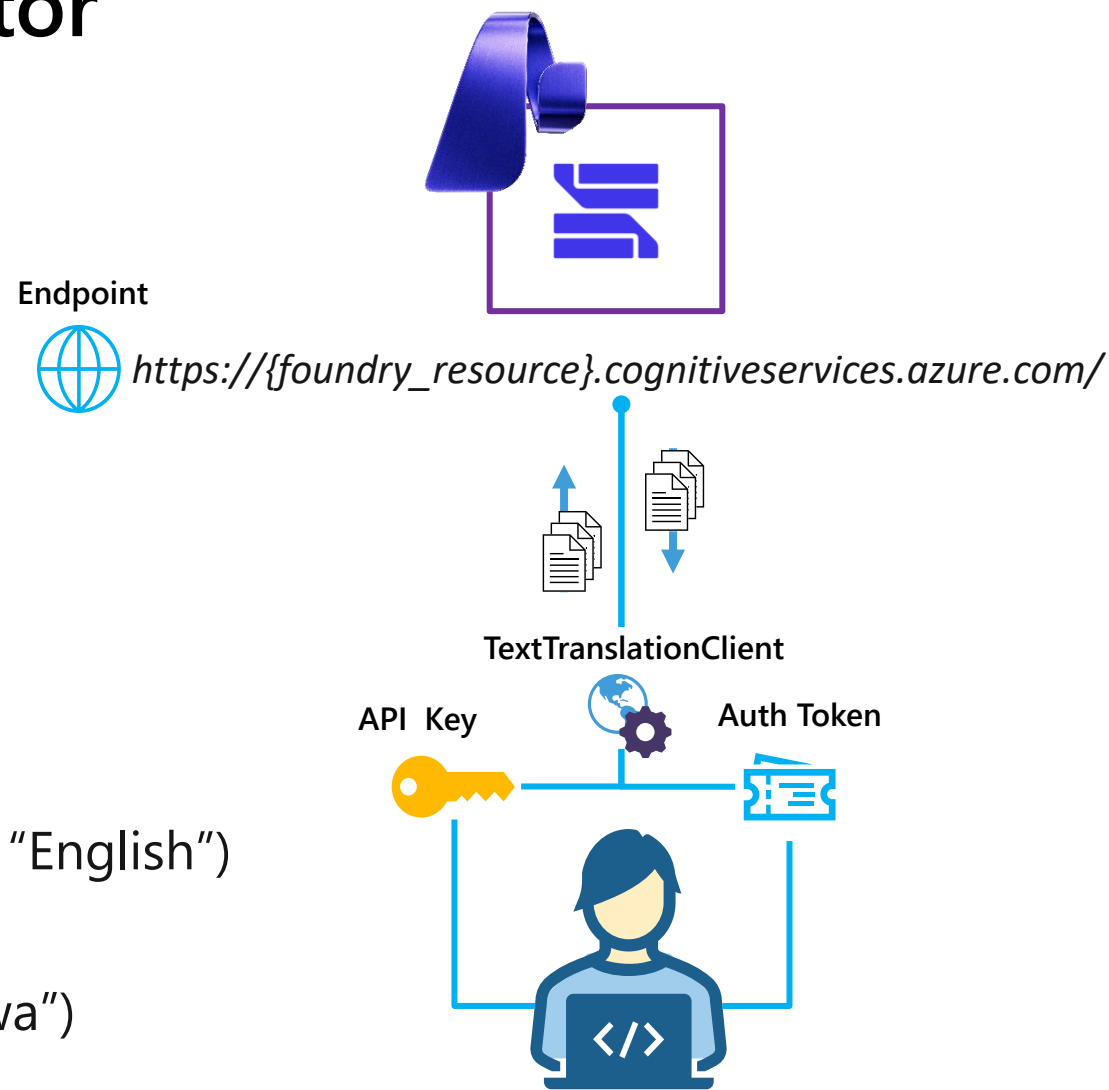
Translate text with Azure Translator

1. Connect to the API

- **Endpoint:**
 - Microsoft Foundry resource
 - Global / Regional translator endpoint
- **Credentials:**
 - Microsoft Foundry project API key
 - Authentication token

2. Use a *TextTranslationClient* to perform tasks

- Get supported languages (for example "en" : "English")
- Translate text ("こんにちは" = "Hello")
- Transliterate text ("こんにちは" = "Kon'nichiwa")



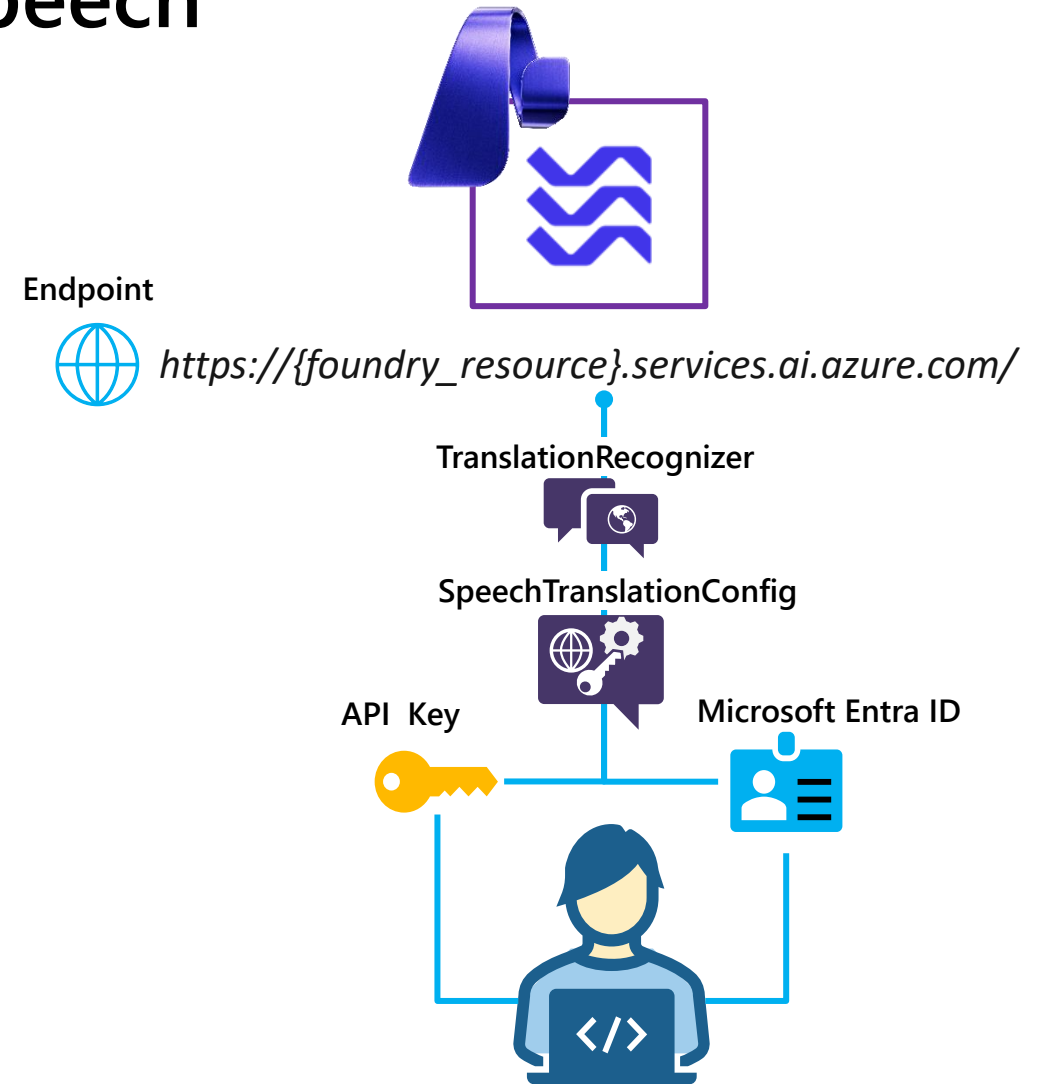
Translating speech with Azure Speech

1. Connect to the API

- **Endpoint:**
 - Microsoft Foundry resource*
 - Global/Regional endpoint
- **Credentials:**
 - Microsoft Foundry project API key
 - Microsoft Entra ID

2. Create a *SpeechTranslationConfig* object for the connection

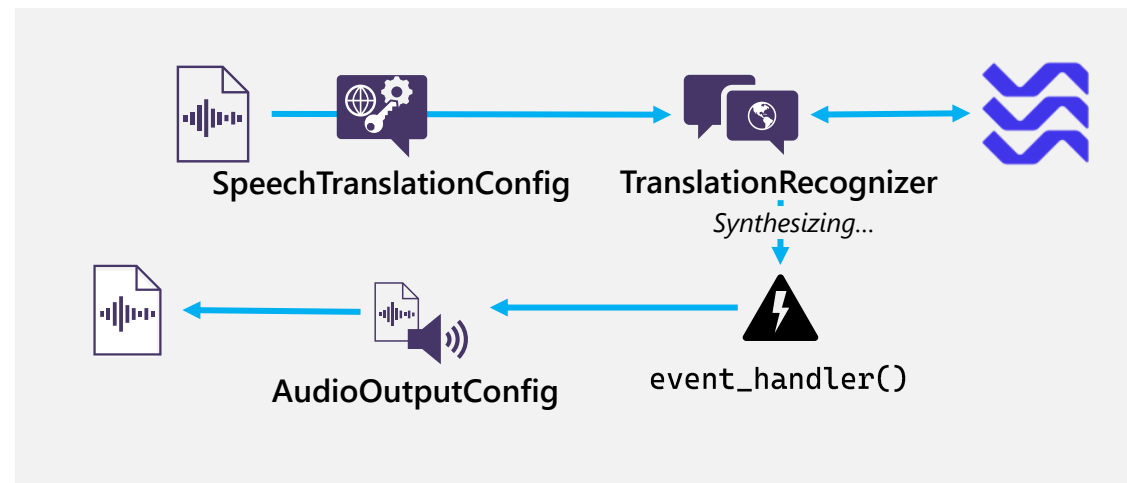
3. Use the *TranslationRecognizer* object for translation operations



Synthesizing Speech Translations

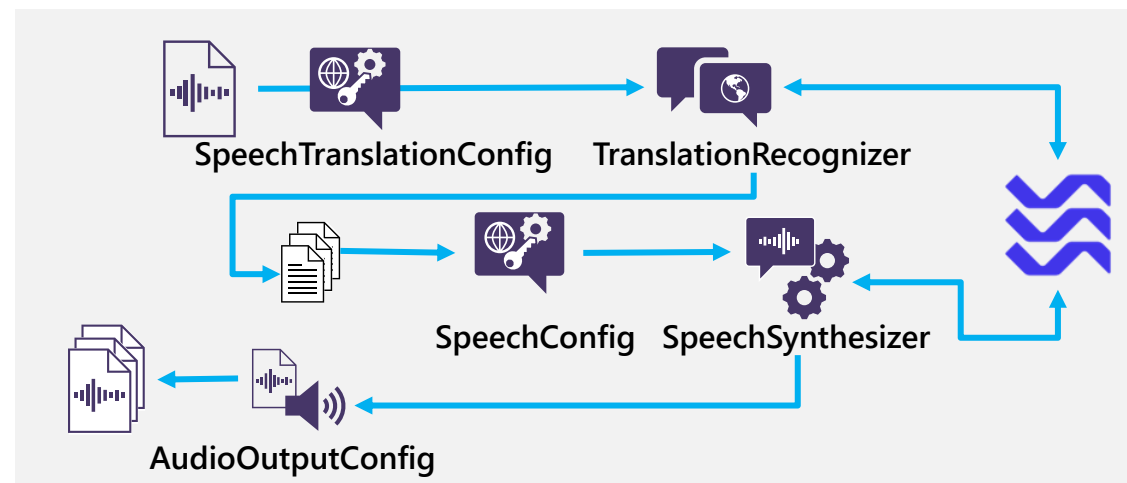
Event-based synthesis

- Only supported for 1:1 translation (single target language)
- Specify desired voice in the **TranslationConfig**
- Use the *Synthesizing* event to retrieve audio stream
- Create an event handler and use *Result.GetAudio()* to retrieve byte stream



Manual synthesis

- Use for multiple target languages
- Translate to text then use Text-to-Speech API to synthesize each translation in the results



Exercise

Translate text and speech

In this exercise, you'll:

- Use Azure Translator in Foundry Tools to translate text
- Use Azure Speech in Foundry Tools to translate speech

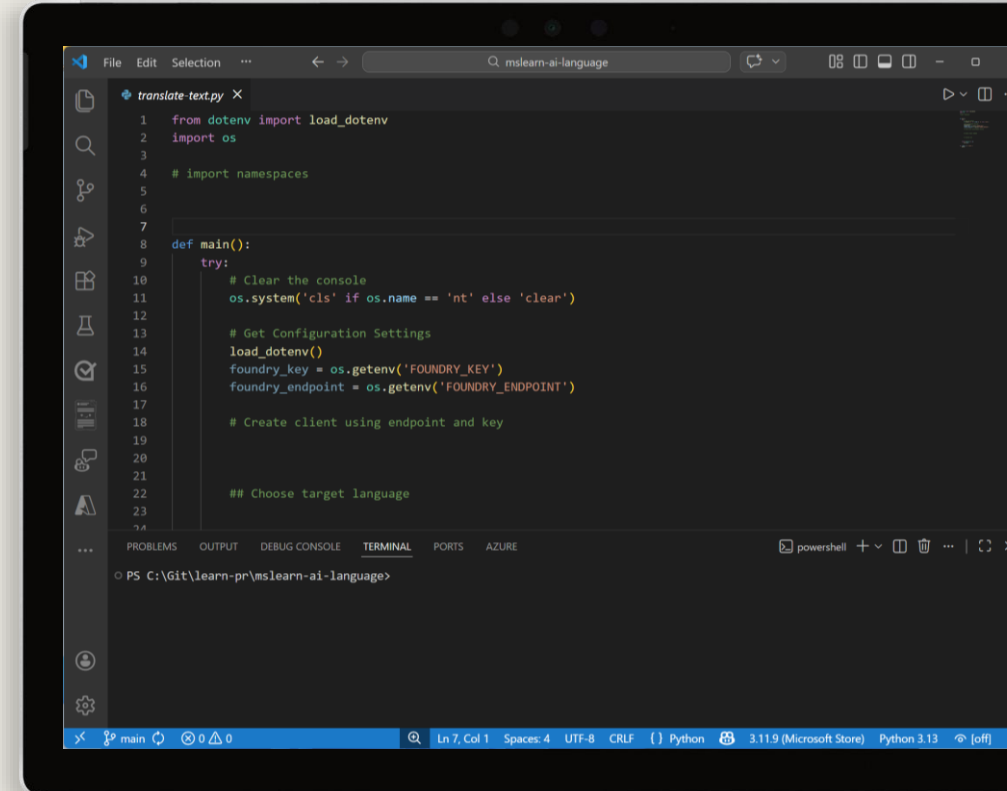
Start the exercise at:

19. Translate text and speech



1 h

0%



Knowledge check



- 1** What function of an Azure Translator TextTranslationClient object should you use to convert the Chinese word "你好" to the English word "Hello"?
 - ☐ get_supported_language
 - ☒ translate
 - ☐ transliterate

- 2** What function of an Azure Translator TextTranslationClient object should you use to convert the Russian word "спасибо" in Cyrillic characters to "spasibo" in Latin characters?
 - ☐ get_supported_language
 - ☐ translate
 - ☒ transliterate

- 3** Which Azure Speech SDK object should you use to specify the language(s) into which you want speech translated?
 - ☐ SpeechConfig
 - ☒ SpeechTranslationConfig
 - ☐ AudioConfig



